

PYEXPO25

Genius innovation leaves behind a legacy...



Team ID: T040

Team Name: Future Forge (Registered Name)

PS Number: PY028

PS Title: Remote Door Lock Control

Domain: IOT

Category: Hardware



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Problem Statement:

Problem Solved:

- **Security Risks:** Prevents unauthorized access and alerts users of suspicious activities.
- **Convenience:** Allows users to control door locks remotely, check lock status, and grant temporary access to guests.
- **Access Management:** Maintains a log of access events for security audits.
- **Power Outage Usability:** Ensures manual override functionality during power failures.

Target Audience:

- **Homeowners:** For enhanced home security and remote access control.
- **Office Managers:** To manage employee access and improve workplace security.
- **Property Managers:** To facilitate temporary access for maintenance staff or guests.
- **Tech-Savvy Users:** Interested in smart home integration and IoT solutions.



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Proposed Solution:

IoT System Description and Purpose:

The IoT-enabled smart door lock system is designed to provide enhanced security and remote access control for homes and offices. Its primary purpose is to allow users to lock or unlock doors remotely, monitor access events, and receive real-time alerts for unauthorized attempts. The system also ensures usability during power outages by supporting manual operation.

Devices and Sensors Involved:

1. Smart Door Lock Hardware:

1. **Electronic Locking Mechanism:** Controls the physical locking and unlocking of the door.
2. **Motor Driver:** Powers the locking mechanism for smooth operation.
3. **Manual Override:** Allows manual key operation during power failures.

2. Sensors:

1. **Biometric Sensor:** Fingerprint or face recognition for secure and convenient authentication.
2. **Magnetic Door Sensor:** Detects the door's open or closed status.
3. **Tamper Sensor:** Alerts users of any unauthorized physical manipulation or break-in attempts.

3. Communication Modules:

1. **Wi-Fi Module:** Enables remote connectivity to the mobile app for monitoring and control.
2. **Bluetooth Module:** Allows local access via mobile devices when near the door.

4. Microcontroller Unit (MCU):

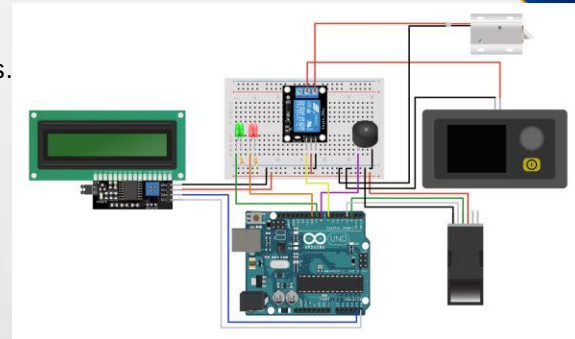
1. Coordinates communication between sensors, locking mechanism, and cloud services.
2. Popular choices: ESP32 or Raspberry Pi for IoT integration.

5. Mobile Application:

1. Provides remote control, access logs, and real-time notifications.
2. Allows generation of one-time access codes for guests.

6. Cloud Server:

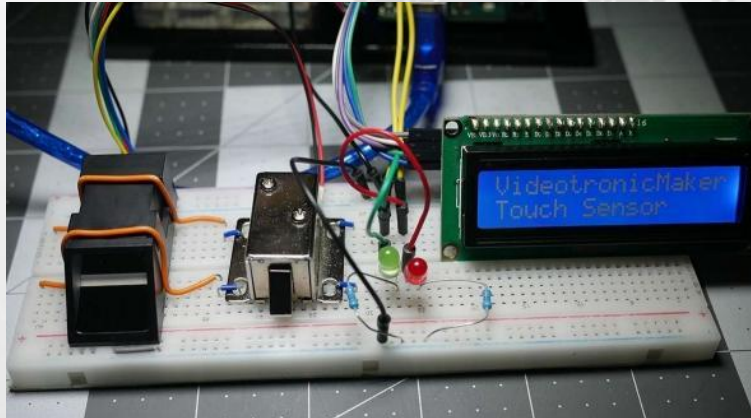
1. Stores access logs securely.
2. Manages user authentication and remote commands.



Architecture & Hardware-Software:

Apps and platforms

- Arduino IDE 2.0 (beta)



components

1 Buzzer, Piezo ☒

1 Resistor 330 ohm ☒

1 Jumper wires (generic) ☒

1 HiLetgo 2pcs 5V One... ☒

1 5mm Green LED ☒

1 Optical Fingerprint... ☒

1 5mm Red LED ☒

1 ATOPLEE Electromagn... ☒

1 Arduino UNO Wifi Rev.2 ☒



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Demo:

Add screenshot



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Impact and Future Scope:

- Actually , It will help our users to take care of the things even babies to not go out of the room
- Also it will be usefull for largescale industries to maintain a valuable things within a safety room
- On other hand , we can add some new features to thr door lock system a , it is customizeable

Research and References:

- <https://projecthub.arduino.cc/Tishin/arduino-fingerprint-sensor-solenoid-lock-wlcd-f01a40>



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THANK YOU !

